

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No - 10	Team No - 15	Project Title - Hospital Management & Appointment System
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1. Abstract

The Hospital Management & Appointment System is a web-based application developed to make hospital services more organized, accessible, and efficient for both patients and hospital staff. In many hospitals, important activities such as patient registration, appointment booking, medical records, billing, pharmacy management, and laboratory reports are handled through separate systems or manual processes. This can cause delays, long waiting times, data errors, difficulty in accessing patient information, and poor coordination between different departments. To overcome these problems, the proposed system brings all major hospital operations together on a single integrated platform. Patients can register, maintain their medical information, book appointments, and check their appointment status. Doctors can manage their schedules and access Electronic Medical Records (EMR) to understand a patient's medical history. The system also includes token and queue management to reduce waiting time, billing and insurance management for smoother financial processes, pharmacy and inventory management to track medicines, and a laboratory module for managing and accessing test reports. An admin dashboard allows authorized staff to monitor and manage overall hospital activities. The system will be developed using React.js for the frontend, Node.js and Express.js for the backend, MySQL with Sequelize for database management, JWT for secure authentication, and Twilio for communication and notifications. Overall, the project aims to reduce paperwork, improve coordination, save time, minimize errors, and provide a simple and convenient hospital experience for patients, doctors, and administrators.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

- **Frontend:** React.js
- **Backend:** Node.js with Express.js
- **Database:** MySQL (RDBMS)
- **ORM:** Sequelize
- **Authentication:** JWT (JSON Web Token)
- **API:** RESTful APIs using Node.js/Express
- **Database Concepts:** SQL, ER Modelling, Normalisation, Constraints, Joins, Transactions, Views and Stored Logic
- **Security:** JWT-based authentication and role-based access control
- **Communication:** Twilio for SMS/notification services
- **Testing:** Node.js testing tools
- **Development Tools:** Visual Studio Code, Git and GitHub
- **Deployment/Containerisation:** Docker (if required)

Guide Name and Signature (remarks if any) _____